

WORKSHOP SAFETY, HEALTH, AND ENVIRONMENT

ME1031 – Engineering Skill Development

Department of Mechanical Engineering, Sri Lanka
Institute of Information Technology



GIF courtesy: <https://gifer.com/en/JHWh>

Presented by Dr. L U Subasinghe | PhD (SG), BSc Eng Hons (SL), CEng.MIE(SL), AMIMechE
Senior Lecturer

Department of Mechanical Engineering, University of Moratuwa
lihils@uom.lk

LEARNING OUTCOMES

- After completing this session, the students should be able to,
 - LO-5: Practice safety guidelines and wearing of personal protective equipment in the engineering workshop

OUTLINE

- Introduction to workshop safety, health, and environment
- Incidents and accidents
- Unsafe conditions and acts
- Hazards and risks
- Risk control
- Accidents happened due to unsafe practices
- Fire safety

INTRODUCTION TO WORKSHOP SAFETY, HEALTH, AND ENVIRONMENT

WHAT IS WORKSHOP SAFETY, HEALTH, AND ENVIRONMENT?

- Safety
 - The **protection from** the risk of **harm or injury to humans** or **damage to machines and tools** in the workshop
- Health
 - The promotion and maintenance of the highest degree of **physical, mental** and **social well-being** of people in the workshop
- Environment
 - Keeping the surroundings of the workshop **clean and tidy**

LEGISLATION

- Factories Ordinance (1942 No.45)
 - An ordinance to make provision for the safety and welfare of workers in factories
 - Consists of 15 subsections
 - Part 1: Registration of factories and approval of factory buildings
 - Part II: Health (General provision)
 - Part III: Safety (General provision)
 - Part IV: Welfare (General provision)
 - Part V: Health, safety and welfare (Special provision and regulations)
 - Part VI: Notification and investigation of accidents and industrial diseases
 - Part VII: Employment of women and young persons- hours and holidays
 - Part VIII: Special applications and extensions- Premises in respect of which owner is liable
 - Part IX: Home work
 - Part X: Miscellaneous
 - Part XI: Administration
 - Part XII: Repealed by Law No 12-1976
 - Part XIII: Supplementary- offences, penalties and legal proceedings
 - Part XIV: Application of ordinance
 - Part XV: Interpretation and general interpretation

INCIDENTS AND ACCIDENTS

WHAT IS AN INCIDENT?

- **An incident** is an event that does not result in an injury or illness to the workers or damages to the properties
 - Also includes **near-misses**
- A **near miss, near hit or close call** is an unplanned event that has the potential to cause, but does not actually result in human injury, environmental or equipment damage, or an interruption to normal operation



Occupational Health & Safety Management System by Eng. Luxman Graciyas Jansz at Workplace Safety for Engineers (Live CPD/July 2021)
<https://www.oshatrain.org/courses/mods/700m6.html>
[https://en.wikipedia.org/wiki/Near_miss_\(safety\)](https://en.wikipedia.org/wiki/Near_miss_(safety))

Image courtesy: Internet

WHAT IS AN ACCIDENT?

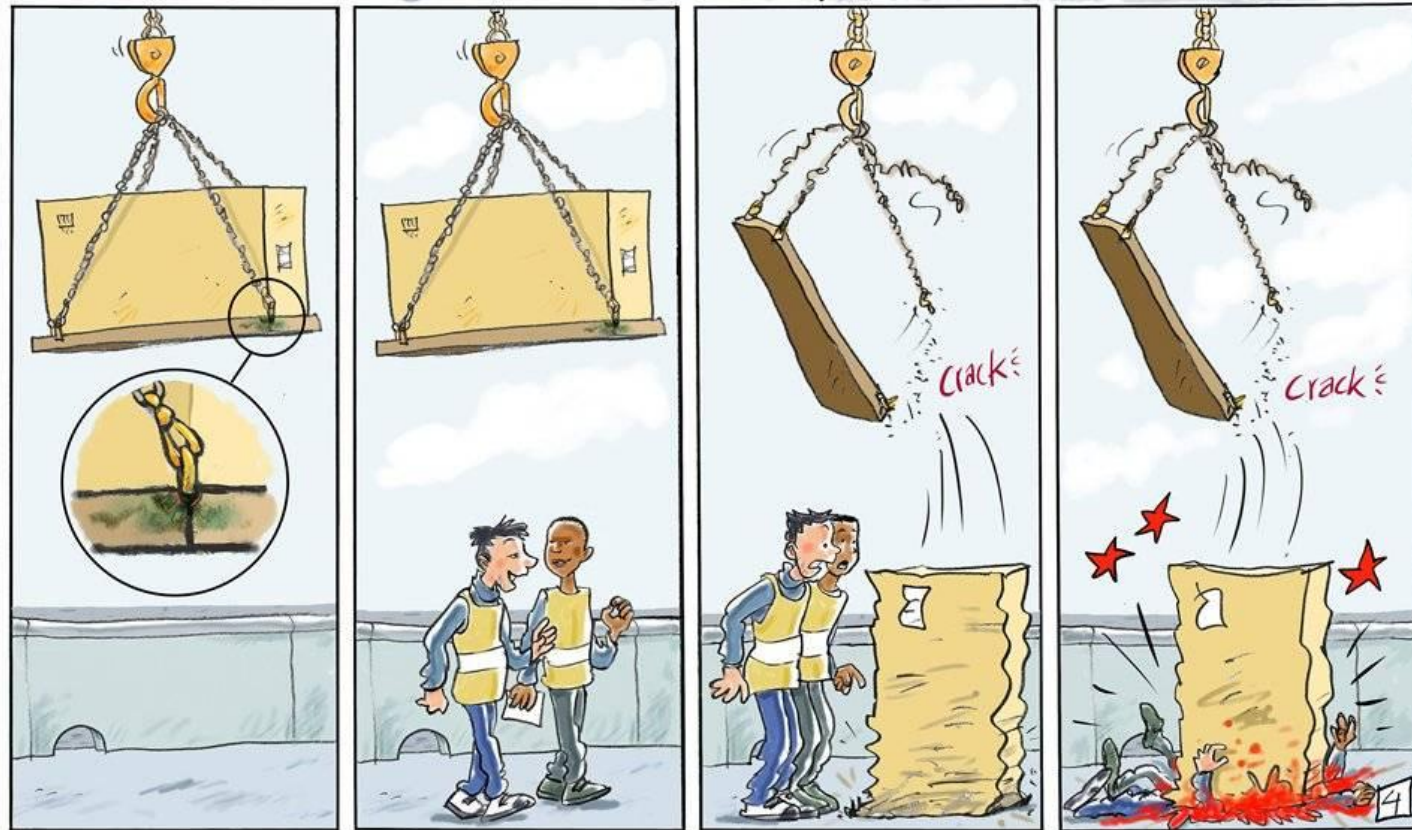
- An accident is an unexpected, unplanned event that results in death, injury or illness to the workers or damages to the properties
 - Accidents are also called **mishaps**



UNSAFE CONDITIONS AND ACTS

UNSAFE CONDITIONS AND UNSAFE ACTS

the rusty lifting gear



Unsafe condition.

Unsafe act

Near miss

Accident

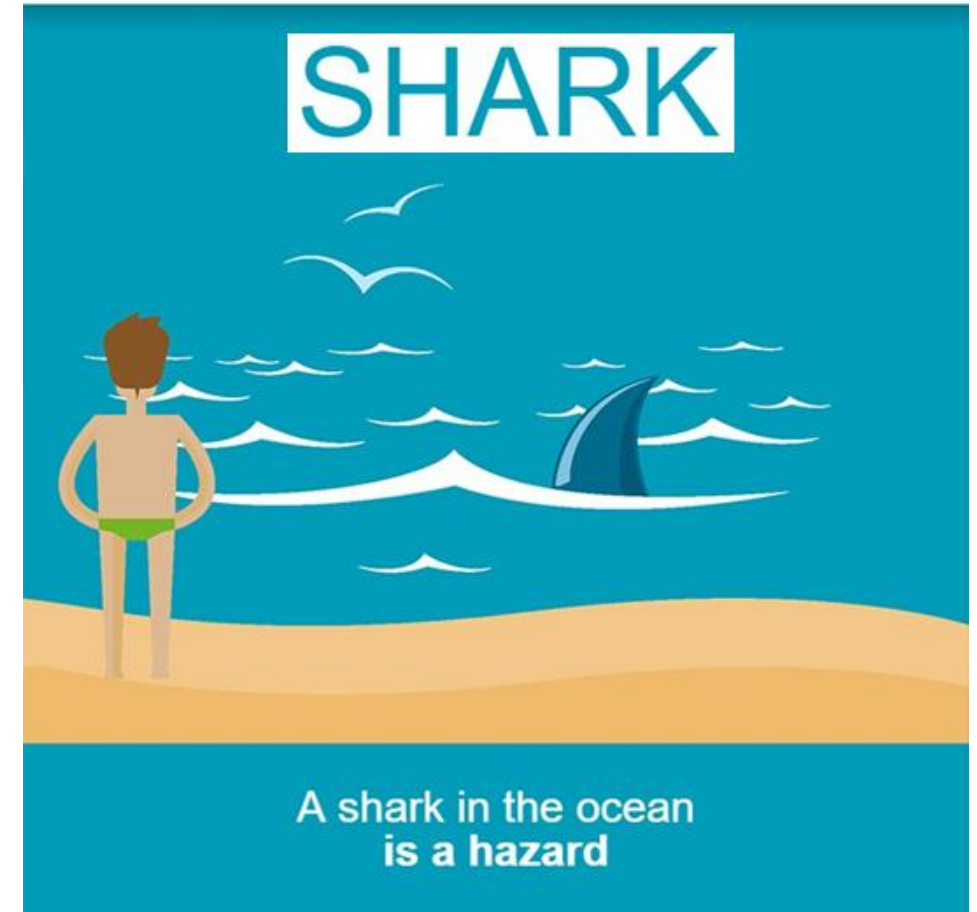
HAZARDS AND RISKS

WHAT IS A HAZARD?

- Source or situation with a potential for harm in terms of
 - human injury or ill health
 - damage to property
 - damage to workplace environment or
 - a combination of above

Hazard

A hazard is something that has the **potential to cause harm**



WHAT IS A RISK?

- A risk is the chance (probability), high or low, that any hazard will actually cause somebody harm
- Can be assessed by determining the severity (consequences) and likelihood of a specific hazardous event

$$\text{Risk} = \text{Severity} * \text{Likelihood}$$

Risk

Risk is the **probability** that a hazard **will cause harm**



DIFFERENT TYPES OF HAZARDS



SIX TYPES OF INDUSTRIAL HAZARDS

1. Safety hazards
2. Physical hazards
3. Chemical hazards
4. Biological hazards
5. Ergonomic hazards
6. Psychological / Workload hazards

A FEW EXAMPLES



Safety Hazards



Biological Hazards



Physical Hazards



Ergonomic Hazards



Chemical Hazards

©2019 Creative Safety Supply

HOW TO DETERMINE THE RISK?

Likelihood Categories		
(1)	Rare	Once every 5 years
(2)	Unlikely	Once per year
(3)	Moderate likely	Once per month
(4)	Likely	Once per week
(5)	Almost certain	Once per day

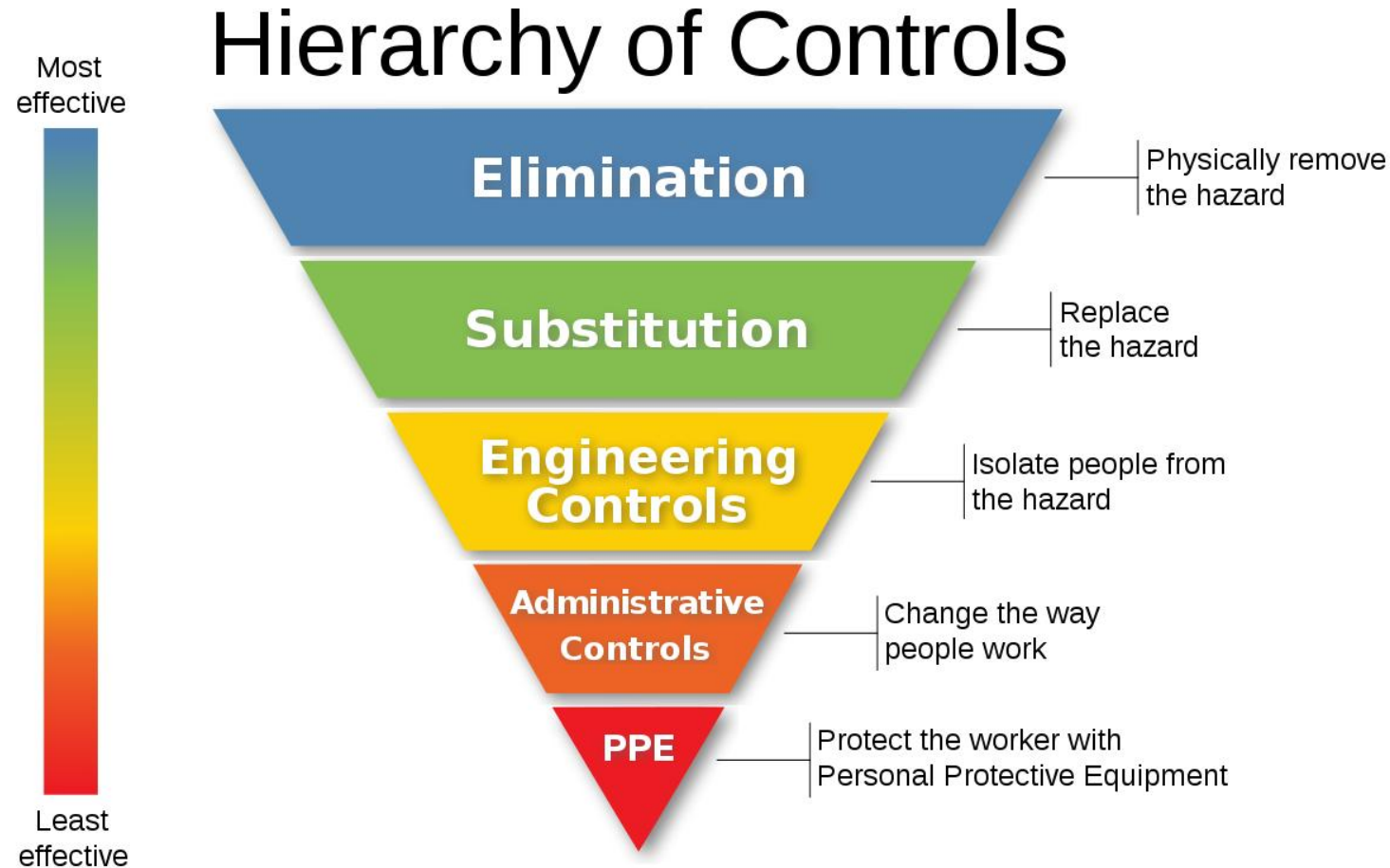
Severity Categories		
(1)	Insignificant	No impact or not detectable
(2)	Minor	Impact to compliance
(3)	Moderate	Impact to aesthetic
(4)	Major	Impact to regulatory
(5)	Catastrophic	Impact to public or human health

Likelihood	Severity of Consequences					Cut-Off Point and Meaning of the Color
	Insignificant	Minor	Moderate	Major	Catastrophic	
Almost certain	5	10	15	20	25	1-5: low risks (green)
Likely	4	8	12	16	20	6-9: medium risks (yellow)
Moderate likely	3	6	9	12	15	10-15: high risks (orange)
Unlikely	2	4	6	8	10	≥16: very high risks (red)
Rare	1	2	3	4	5	

RISK CONTROL

RISK CONTROL

- Risk control is a part of the risk management process in which methods for neutralising or reduction of identified risks are implemented



PERSONAL PROTECTIVE EQUIPMENT

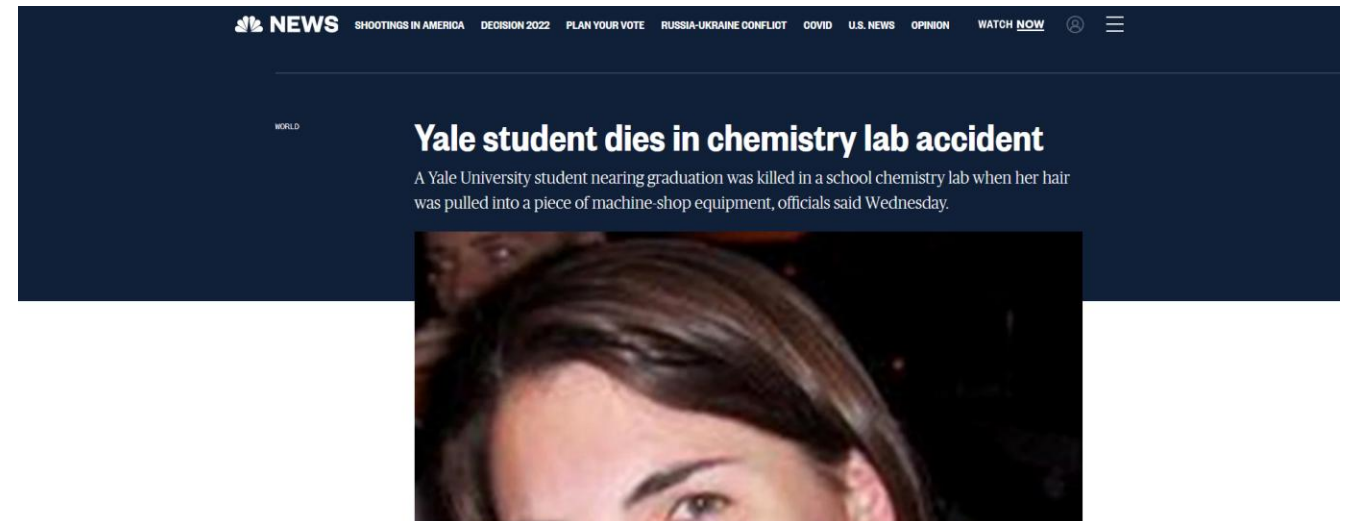
- Equipment worn by individuals which work as the “last line of defense” to protect workers from injuries
- Also known as PPE



ACCIDENTS HAPPENED DUE TO UNSAFE PRACTICES

YALE STUDENT DIES IN CHEMISTRY LAB ACCIDENT

- Date: 12th April 2011
- **Her hair** was pulled into a fast-spinning lathe machine
 - Neck compressed → Died from asphyxiation
- Analysis
 - She was trained to use equipment
 - She was working alone
 - She was working at night-time with limited people to help
 - **She did not tie her hair up**



INCIDENT SHOWS DANGER OF LOOSE CLOTHING

- Date: February 2009
- A car wash employee was nearly strangled when **her scarf** got caught in a rotating scrubber brush
 - Neck compressed
 - Luckily, rescued by a car wash customer
 - He jumped out of his car as it was going through the car wash, cut her free using a pocket-knife, and performed mouth-to-mouth to restore her breathing
- Analysis
 - She was trained to operate the scrubber brush
 - **She was wearing a loose scarf (loose clothing)**



<https://www.safetynewsalert.com/incident-shows-danger-of-loose-clothing/>

A FEW MORE ACCIDENTS CAUGHT ON CAMERA

- <https://www.youtube.com/watch?v=BNhk4NgrFKI>
- <https://www.youtube.com/watch?v=0PzPfzLeDa0>
- <https://www.youtube.com/watch?v=tKs9NmkxzLw>

FIRE SAFETY

CLASSES OF FIRE AND SUITABLE FIRE EXTINGUISHERS

Fire Extinguisher Type	CLASS A	CLASS B	CLASS C	CLASS D	Electrical	CLASS F	Comments
	Combustible materials (e.g. paper & wood)	Flammable liquids (e.g. paint & petrol)	Flammable gases (e.g. butane and methane)	Flammable metals (e.g. lithium & potassium)	Electrical equipment (e.g. computers & generators)	Deep fat fryers (e.g. chip pans)	
Water	✓	✗	✗	✗	✗	✗	Do not use on liquid or electric fires
Foam	✓	✓	✗	✗	✗	✗	Not suited to domestic use
Dry Powder	✓	✓	✓	✓	✓	✗	Can be used safely up to 1000 volts
CO2	✗	✓	✗	✗	✓	✗	Safe on both high and low voltage
Wet Chemical	✓	✗	✗	✗	✗	✓	Use on extremely high temperatures

- Multipurpose fire extinguishers are also available
 - Class AC, Class BC, Class ABC

DIFFERENT TYPES OF FIRE EXTINGUISHERS



HOW TO USE A FIRE EXTINGUISHER

- **P.A.S.S.** technique:

- **P** - Pull the pin on the fire extinguisher in order to break the tamper seal
- **A** - Aim the fire extinguisher low, with the nozzle pointed at the base of the fire
- **S** - Squeeze the handle of the fire extinguisher to release the extinguishing agent
- **S** - Sweep the nozzle from side to side while pointed at the base of the fire until it is extinguished
 - If the fire re-ignites, repeat the last 3 steps



https://ohsonline.com/-/media/OHS/OHS/Images/2004/08/abc_fire_extinguisher.jpg
<https://www.youtube.com/watch?v=ktrv34zW7-A>
<https://www.youtube.com/watch?v=PQV71INDaY>

Image courtesy: Internet



THE END